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MAE 3270 HW 9 Q3

Prof Alan Zehnder Michelle Zhou Oct 24 & 29, 2021 With the help of Oct 18 Lecture Notes

```
close all
clc; clear;
```

Overlayed graph

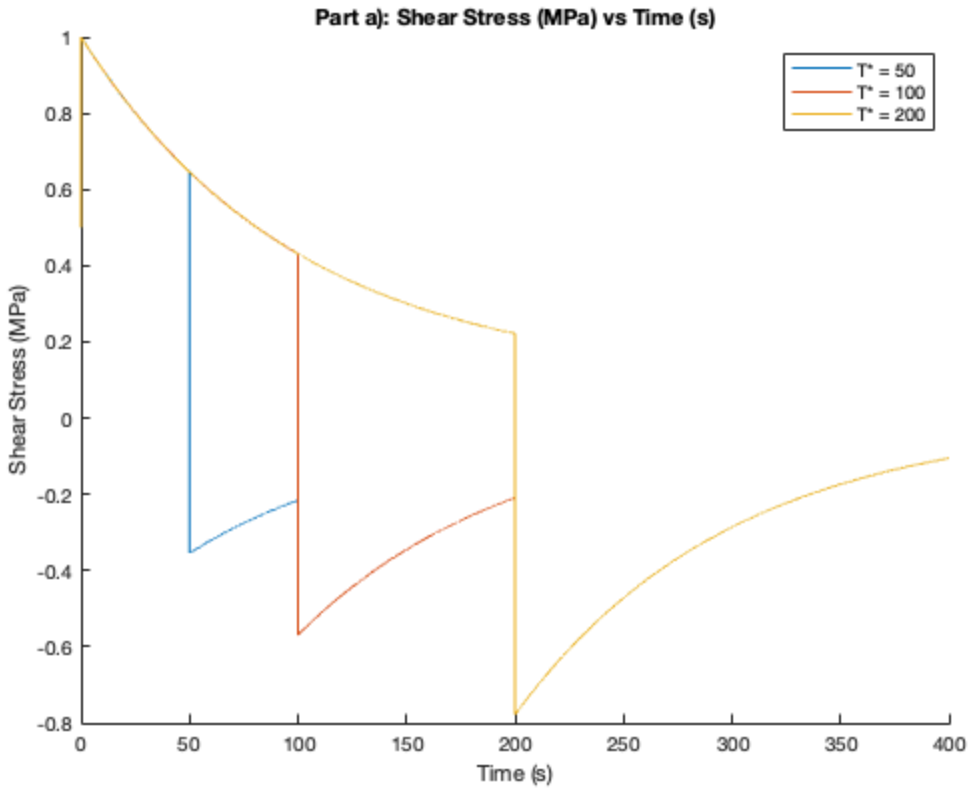
```
T_list = [50 100 200];
for i = 1:3
    hold on
    % create time array
    T = T_list(i);
    t = linspace(0,2*T, 100*T);

    % shear modulus
    G_inf = 100*10^6;           %% Pa
    G_1 = 900*10^6;             %% Pa
    tau = 100;                  %% Pa
    G = G_inf + G_1*exp(-t/tau);
    G_shift = G_inf + G_1*exp(-(t-T)/tau);

    % heaviside step function
    shear_stress = 0.001*(heaviside(t).*G - heaviside(t-T).*G_shift);

    % part a): plot shear stress vs time
    plot(t, shear_stress*10^-6)

end
xlabel('Time (s)')
ylabel('Shear Stress (MPa)')
title('Part a): Shear Stress (MPa) vs Time (s)')
legend('T* = 50', 'T* = 100', 'T* = 200')
```



Separate graphs

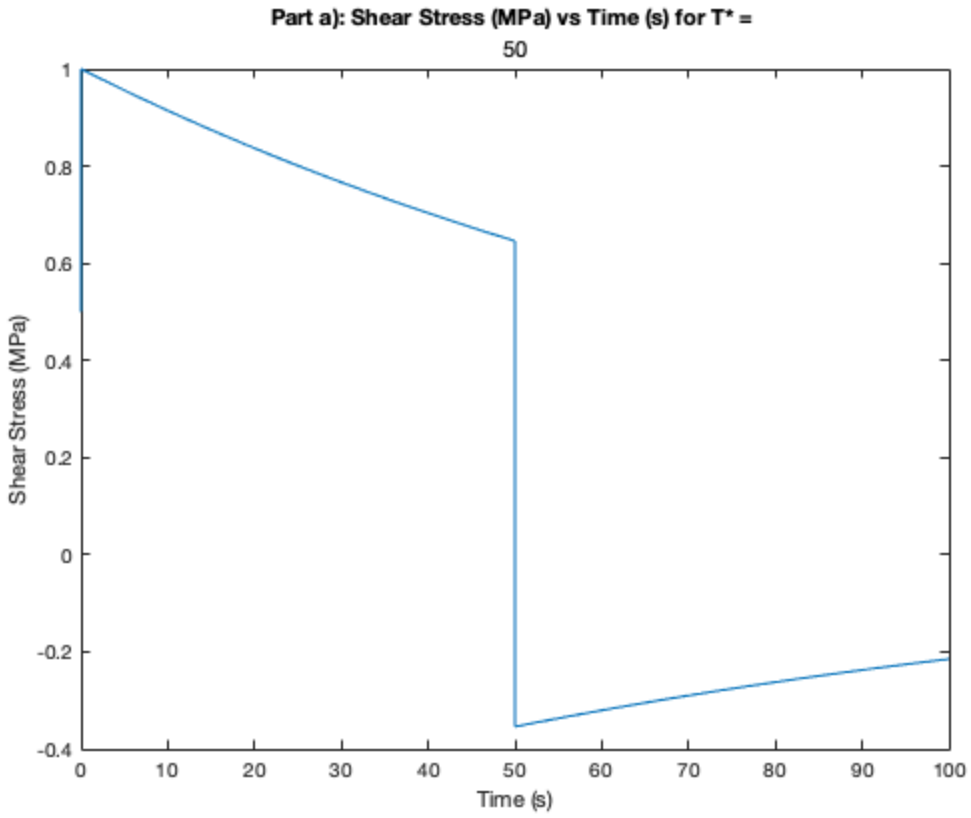
```
for i = 1:3
    % create time array
    T = T_list(i);
    t = linspace(0,2*T,10000*T);

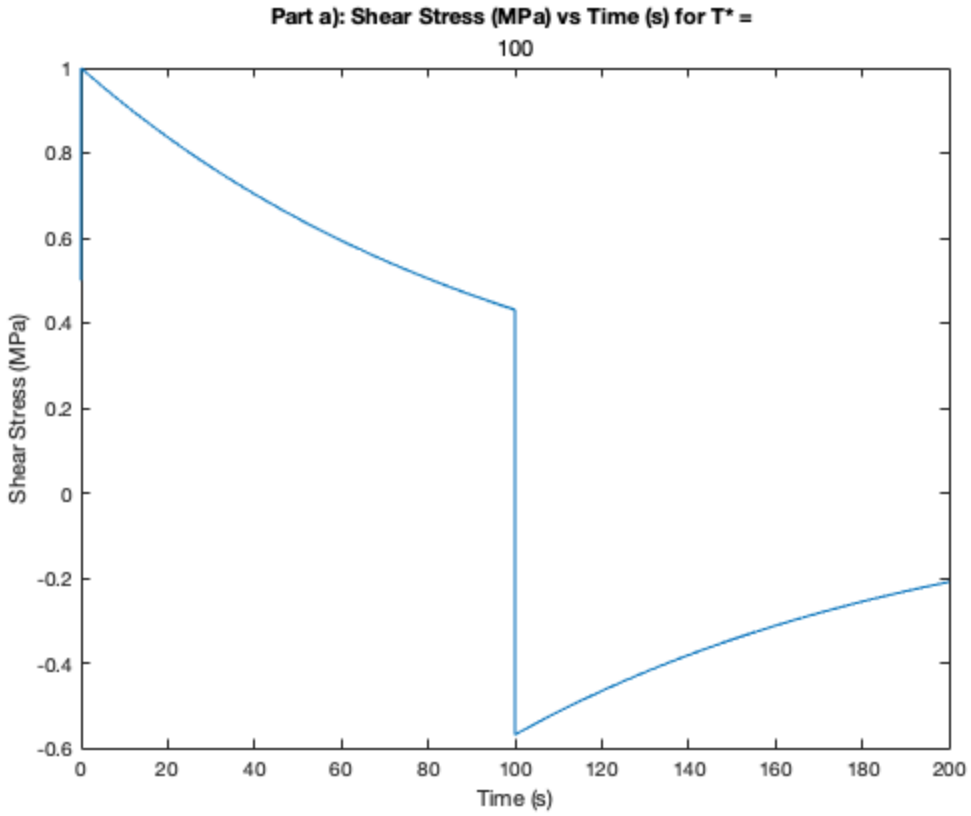
    % shear modulus
    G_inf = 100*10^6;           %% Pa
    G_1 = 900*10^6;             %% Pa
    tau = 100;                  %% Pa
    G = G_inf + G_1*exp(-t/tau);
    G_shift = G_inf + G_1*exp(-(t-T)/tau);

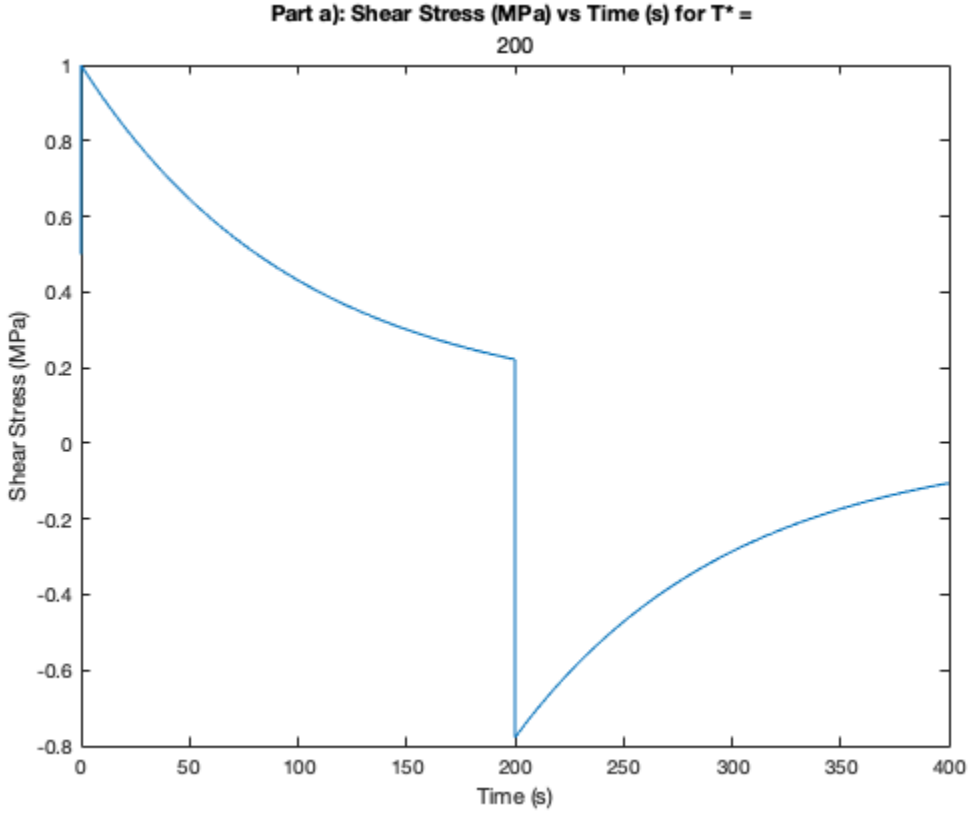
    % shear stress
    shear_stress = 0.001*(heaviside(t).*G - heaviside(t-T).*G_shift);

    % part a): plot shear stress vs time
    figure
    plot(t, shear_stress*10^-6)
    xlabel('Time (s)')
    ylabel('Shear Stress (MPa)')
    title('Part a): Shear Stress (MPa) vs Time (s) for T* = ', T)
```

```
end
```


Part a): Shear Stress (MPa) vs Time (s) for T* =
50
Shear Stress (MPa)
Time (s)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
0
10
20
30
40
50
60
70
80
90
100


Part a): Shear Stress (MPa) vs Time (s) for T* =
100
Shear Stress (MPa)
Time (s)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
20
40
60
80
100
120
140
160
180
200


Part a): Shear Stress (MPa) vs Time (s) for T* =
200
Shear Stress (MPa)
Time (s)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
50
100
150
200
250
300
350
400

Part b) plot shear stress vs shear strain for T* = 100s

```
% time
T = 100;
t = linspace(-10,2*T,100000);

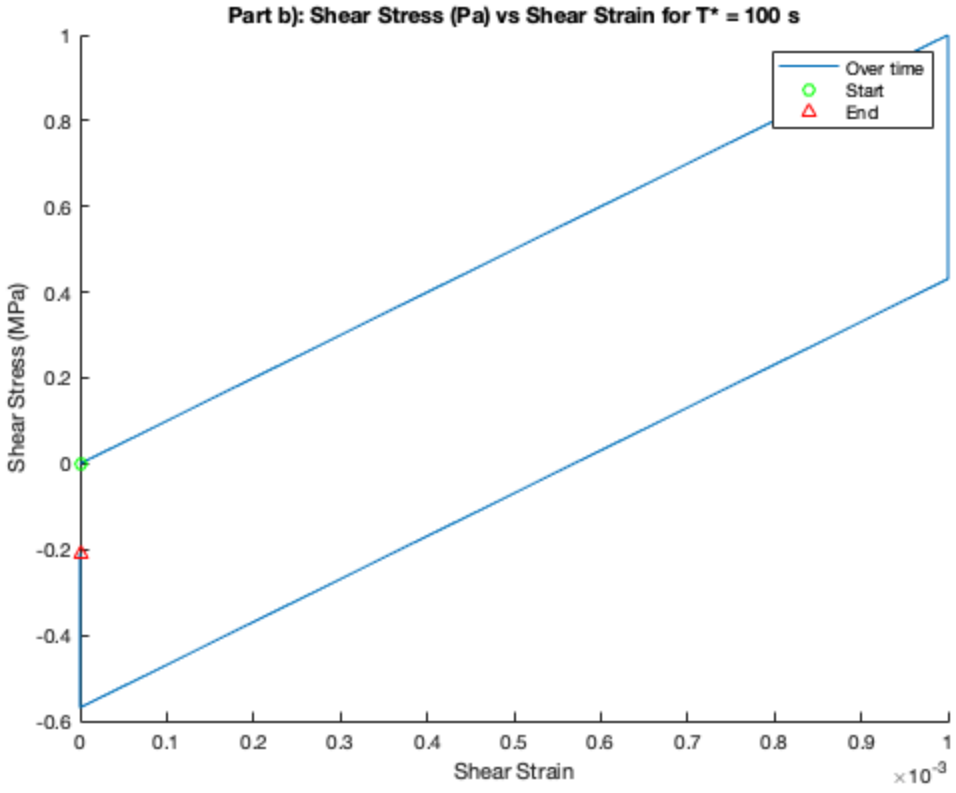
% shear modulus
G = G_inf + G_1*exp(-t/tau);
G_shift = G_inf + G_1*exp(-(t-T)/tau);

% shear strain
gamma = 0.001*(heaviside(t) - heaviside(t-T));

% shear stress
shear_stress = 0.001*(heaviside(t).*G - heaviside(t-T).*G_shift);

figure
hold on
plot(gamma, shear_stress*10^-6)
plot(gamma(1), shear_stress(1)*10^-6, 'go')

plot(gamma(length(gamma)), shear_stress(length(gamma))*10^-6, 'r^')
xlabel('Shear Strain')
ylabel('Shear Stress (MPa)')
title('Part b): Shear Stress (Pa) vs Shear Strain for T* = 100 s')
legend('Over time', 'Start', 'End')
```



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